

Observability in Semantic Kernel

Brief introduction to observability

When you build AI solutions, you want to be able to observe the behavior of your services. Observability is the ability to monitor and analyze the internal state of components within a distributed system. It is a key requirement for building enterprise-ready AI solutions.

Observability is typically achieved through logging, metrics, and tracing. They are often referred to as the three pillars of observability. You will also hear the term "telemetry" used to describe the data collected by these three pillars. Unlike debugging, observability provides an ongoing overview of the system's health and performance.

Useful materials for further reading:

- [Observability defined by Cloud Native Computing Foundation](#)
- [Distributed tracing](#)
- [Observability in .Net](#)
- [OpenTelemetry](#)

Observability in Semantic Kernel

Semantic Kernel is designed to be observable. It emits logs, metrics, and traces that are compatible to the OpenTelemetry standard. You can use your favorite observability tools to monitor and analyze the behavior of your services built on Semantic Kernel.

Specifically, Semantic Kernel provides the following observability features:

- **Logging:** Semantic Kernel logs meaningful events and errors from the kernel, kernel plugins and functions, as well as the AI connectors.

❗ Important

Traces in Application Insights represent traditional log entries and OpenTelemetry span events . They are not the same as distributed traces.

- Run

Time range: Last 7 days

Limit: 1000

KQL mode

```

1 customMetrics
2 | where name startswith "semantic_kernel.function"
3 | project timestamp, name, customDimensions, value
4 | take 10

```

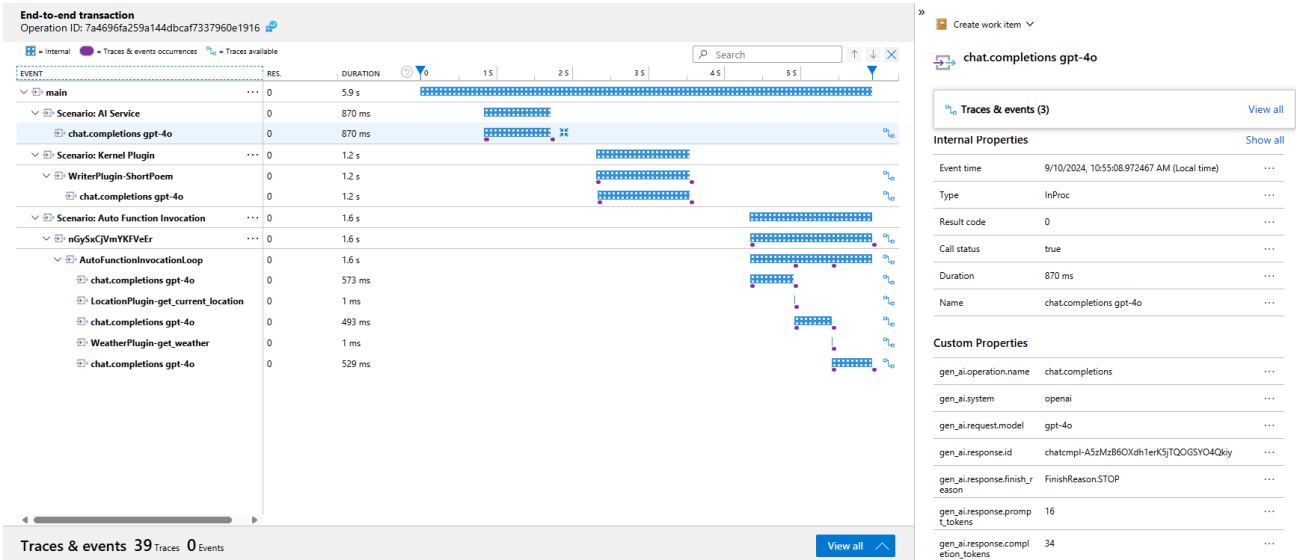
Results

Chart

timestamp [UTC] ↑↓	name	customDimensions	value
9/10/2024, 5:48:18.388 PM	semantic_kernel.function.invocation.duration	{semantic_kernel.function.name:"roMWntlbjukOoeSb"}	8.93088910001097
timestamp [UTC]	2024-09-10T17:48:18.388407Z		
name	semantic_kernel.function.invocation.duration		
customDimensions	{semantic_kernel.function.name:"roMWntlbjukOoeSb"}		
semantic_kernel.function.name	roMWntlbjukOoeSb		
value	8.93088910001097		
> 9/10/2024, 5:48:18.388 PM	semantic_kernel.function.invocation.duration	{semantic_kernel.function.name:"WeatherPlugin-get_weather"}	0.00181909999810159
> 9/10/2024, 5:48:14.096 PM	semantic_kernel.function.invocation.duration	{semantic_kernel.function.name:"LocationPlugin-get_current_location"}	0.00084900000365451
> 9/10/2024, 5:48:08.780 PM	semantic_kernel.function.invocation.duration	{semantic_kernel.function.name:"WriterPlugin-ShortPoem"}	5.06012109998846
> 9/10/2024, 5:34:05.520 PM	semantic_kernel.function.invocation.duration	{semantic_kernel.function.name:"WeatherPlugin-get_weather"}	0.000147000013384968
> 9/10/2024, 5:34:05.520 PM	semantic_kernel.function.invocation.duration	{semantic_kernel.function.name:"WPjspTMArscxqmHi"}	1.77789489997667
> 9/10/2024, 5:34:04.647 PM	semantic_kernel.function.invocation.duration	{semantic_kernel.function.name:"LocationPlugin-get_current_location"}	0.000246500014327466
> 9/10/2024, 5:34:04.647 PM	semantic_kernel.function.invocation.duration	{semantic_kernel.function.name:"WriterPlugin-ShortPoem"}	1.17128710000543
> 9/10/2024, 5:31:49.523 PM	semantic_kernel.function.invocation.duration	{semantic_kernel.function.name:"WriterPlugin-ShortPoem","error.type":"FunctionExecutionException"}	0.0211023000010755
> 9/10/2024, 5:31:49.523 PM	semantic_kernel.function.invocation.duration	{semantic_kernel.function.name:"aciukltvDeQzMVoA","error.type":"FunctionExecutionException"}	0.00452229997608811

Columns

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Expand table

Telemetry	Description
Log	Logs are recorded throughout the Kernel. For more information on Logging in .Net, please refer to this document . Sensitive data, such as kernel function arguments and results, are logged at the trace level. Please refer to this table for more information on log levels.
Activity	Each kernel function execution and each call to an AI model are recorded as an activity. All activities are generated by an activity source named "Microsoft.SemanticKernel".
Metric	Semantic Kernel captures the following metrics from kernel functions: <code>semantic_kernel.function.invocation.duration</code> (Histogram) - function execution time (in seconds) <code>semantic_kernel.function.streaming.duration</code> (Histogram) - function streaming execution time (in seconds) <code>semantic_kernel.function.invocation.token_usage.prompt</code> (Histogram) - number of prompt token usage (only for <code>KernelFunctionFromPrompt</code>) <code>semantic_kernel.function.invocation.token_usage.completion</code> (Histogram) - number of completion token usage (only for <code>KernelFunctionFromPrompt</code>)

OpenTelemetry Semantic Convention

Semantic Kernel follows the [OpenTelemetry Semantic Convention](#) for Observability. This means that the logs, metrics, and traces emitted by Semantic Kernel are structured and follow a common schema. This ensures that you can more effectively analyze the telemetry data emitted by Semantic Kernel.

Note

Currently, the [Semantic Conventions for Generative AI](#) are in experimental status. Semantic Kernel strives to follow the OpenTelemetry Semantic Convention as closely as possible, and provide a consistent and meaningful observability experience for AI solutions.

Next steps

Now that you have a basic understanding of observability in Semantic Kernel, you can learn more about how to output telemetry data to the console or use APM tools to visualize and analyze telemetry data.

Console

Last updated on 09/25/2024